P-D

P-D-435D
August 14, 1969
SUPERSEDING
Int. Fed. Spec. P-D-00435C (GSA-FSS)
May 10, 1968 and
Fed. Spec. P-D-435B
September 10, 1963

FEDERAL SPECIFICATION

DISHWASHING COMPOUND, MACHINE, CONTAINING STAIN REMOVER, CHLORINE RELEASING TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a dishwashing compound containing a chlorine release stain remover for use in mechanical dishwashers (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip (Polyolefin).
QQ-S-766 - Steel Plates, Sheets and Strip - Corrosion Resisting.
RR-S-366 - Sieve, Test.
PPP-B-636 - Box, Fiberboard.
PPP-P-45 - Packaging of Bulk Quantities of Soaps Detergents and Related Products (Chip, Powdered or Granular Form) for Domestic and Overseas Shipment.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Federal Hazardous Substances Act

21 CFR 191

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 800-58 - Chemical Analysis of Industrial Metal Cleaning Compositions.
D 2022-64 - Section 50-53 - Sampling and Chemical Analysis of Chlorine Containing Bleaches.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Association of Offical Analytical Chemists Publication:

Official Methods of Analyses - Tenth Edition.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Inc., P. O. Box 540 Benjamin Franklin Station, Washington, D. C. 20044.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies shall be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street N.W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material. The dishwashing and stain removing compound shall be a uniform free-flowing material in the granular or bead form.

3.2 Moisture content. The moisture content of the compound shall not exceed 2.0 percent by weight when tested as specified in 4.5.10.

3.3 Silicates. When tested as specified in 4.5.2, the compound shall contain not less than 11.5 percent by weight of silicates expressed as total silica, SiO_2.

3.4 Phosphates. When tested as specified in 4.5.3, the compound shall contain not less than 24.0 percent by weight of total phosphates expressed as P_2O_5. Not more than 4.0 percent of the total P_2O_5 shall be present as orthophosphate (orthophosphates = .025 x total P_2O_5).

3.5 Total alkalinity. When tested as specified in 4.5.4, the alkalinity, expressed as Na_2O, of the compound shall be not less than 30.0 nor more than 35 percent by weight.

3.6 Solubility. The compound shall be soluble, when tested as specified in 4.5.5. Insoluble matter shall be not more than 2.5 mg. per 10 g. of product.

3.7 Available chlorine. When tested as specified in 4.5.6, the compound shall contain not less than 1.8 percent by weight available chlorine.

3.8 pH. When tested as specified in 4.5.7, a 0.25 percent solution of the compound in distilled water at 25° C. shall have a pH of not less than 12.0 nor more than 13.0.

3.9 Soap. When tested as specified in 4.5.8, the compound shall show no evidence of soap.

3.10 Foam height. Foam height in distilled water shall be not greater than 1.5 centimeters (5 sec. value) when tested as specified in 4.5.9.

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3.11 Excluded materials. The compound shall contain none of the following materials: (a) oxygen releasing compounds; (b) calcium hypochlorite; (c) sodium chloride; (d) calcium chloride; (e) arsenic; (f) lead; (g) mercury; (h) antimony; or any other compound having deleterious effects. The compound shall contain not more than 0.003 percent heavy metals expressed as lead.

3.12 Fineness. The particle size of mechanical mixture of ingredients shall be normally uniform in order to minimize segregation of the ingredients. When tested as specified in 4.5.11, not less than 95.0 percent of the material shall fall within the following range:

Passing	Retained
No. 16	No. 170

3.13 Corrosion. When the compound is tested as specified in 4.5.12. weight changes in the corrosion-resisting steel specimens used shall be not more than 1.2 milligrams loss per square centimeter per hour, and specimens shall show no evidence of pitting, etching or discoloration.

3.14 Workmanship. The material shall be thoroughly mixed and present a minimum of segregation of individual constituents or of lumping or caking. It shall be clean and uniform and free of any defects which may impair its utility.

3.15 Labeling. The labeling of the compound shall comply with applicable requirements of the Federal Hazardous Substances Act.

3.16 Use instructions. Unless otherwise specified in the contract or order, each container shall be durably and legibly labeled with use instructions which shall include the following:

DIRECTIONS FOR USE

For machine washing of china, glasses, and silverware.

Maintain a concentration of 0.25 to 0.30 percent (3-1/3 to 4 ounces per 10 gallons of water at a temperature of 140° to 160° F. Operate the washing cycle for its normal washing period.

Not recommended as a dip destainer for chinaware.

Not recommended for cleaning and destaining coffee urns.

KEEP CONTAINER TIGHTLY CLOSED. STORE IN A COOL DRY PLACE.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling.

4.2.1 Sampling lot. For purposes of sampling, a lot shall consist of not more than 1000 pounds of dishwashing compound prepared from one manufactured batch in a single production and offered for delivery at one time. In case the material is produced by continuous run process, the lot shall consist of material from a continuous run of not more than 24 hours.

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4.2.2 Sampling for inspection of container. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) equal to 2.5 percent defective to verify compliance regarding fill, closure, marking, and other requirements not requiring laboratory tests.

4.2.3 Sampling for lot acceptance. For each inspection lot, a one pound composite sample shall be obtained from the lot by combining equal portions of dishwashing compound selected at random throughout the lot. The number of portions to make up the composite shall be as indicated below. The composite sample shall be prepared and thoroughly mixed in an atmosphere of relative humidity within 5 percent of the relative humidity prevailing at the time and place of weighing. The prepared composite shall be immediately placed in a clean, dry, airtight, and watertight container marked with an indentifying lot number and date. Care shall be taken to prevent contamination or alteration of the composite during sampling, weighing, and storing. The sample unit for moisture shall be a one-ounce specimen. The number of sample units of dishwashing compound shall be as indicated below. Each sample unit shall be placed in an individual one-ounce glass, airtight, and watertight container, covered and sealed to prevent any change in moisture.

Lot size (pounds)	Number of sample units or portions
800 or less	2
801 up to and including 1000	3

4.3 Inspection.

4.3.1 Inspection of containers. Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory marking; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Lot acceptance tests. The sample selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.5. Unless otherwise specified, duplicated determination shall be performed on the composite and the sample for moisture content. Failure of the composite sample or the samples for moisture content to meet the requirements of this specification shall be cause for rejection of the lot. All test reports shall contain the individual values utilized in expressing the final results.

4.4 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item as acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE I. Examination of preparation for delivery

Examine	Defects
Containers	Not as specified.
Materials	Any components missing or damaged.
Quantity	Number of unit packages in shipping containers not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Workmanship	Inadequate application of components such as incomplete closure of containers, containers distorted or dirty.

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4.4.1 Examination for closure of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure shall be in accordance with the appendix of that specification.

4.4.2 Test for water vapor permeability. The sampling and testing for water vapor permeability shall be performed in accordance with the quality assurance provisions of L-P-378.

4.5 Test methods.

4.5.1 Preparation of laboratory samples. Prior to making the required analyses, the sample material shall be thoroughly mixed in a tumbler mixer for 1/2 hour. It is extremely important that all of the ingredients be uniformly distributed in the sample used in several of the required tests.

4.5.2 Silicates. Determine silicates in accordance with sections 18 to 21 of ASTM D 800-58.

4.5.3 Phosphates. Determine phosphates in accordance with sections 24 to 31 of ASTM D 800-58.

4.5.4 Total alkalinity as Na_2O. Determine total alkalinity in accordance with sections 4 to 6 of ASTM D 800-85.

4.5.5 Solubility. Add slowly 10 grams of the product in 1000 ml. of agitated distilled water at 24°C. Stir for three minutes with a magnetic stirrer (maximum speed). Filter through a weighed Gooch crucible, wash the crucible twice with small portion of hot water, then dry to constant weight at 105°C. Report weight of precipitate as milligrams of insoluble matter.

4.5.6 Available chlorine. Determine the available chlorine in accordance with paragraphs 4.153 to 4.155 (arsenious oxide titration) of the Offical Methods of Analysis of the Association of Offical Analytical Chemists. A 2 g. sample shall be used.

4.5.7 pH. Determine the pH of a 0.25 percent solution of the compound in distilled water at 25° ± 1°C. by using a suitable pH meter, previously calibrated, which has an alkali resistant glass electrode.

4.5.8 Qualitative test for soap. Dissolve 2 g. of compound in 100 ml. of distilled water. Filter off insoluble matter. Acidify 5 ml. of the filtrate with glacial acetic acid in a test tube. The appearance of fatty acid precipitate or oily drops indicates the presence of soap.

4.5.9 Foam height test.

4.5.9.1 Equipment. The stirring motor is a standard Hamilton Beach Model 33 motor or equal and stirring blade which has been removed from its stand and equipped with a 250 ml. beaker clamp for use in a ring stand. The test cell is a glass cylinder 7.0 cm. inside diameter and 18.5 cm. high made by sealing one end to provide a flat bottom. The baffle used in a cage constructed of 16 gauge stainless steel having four 1/2 inch wide baffles extending into the test cell toward the center from equidistant points on the circumference. These baffles are held in place by 1/2 inch wire rings having a diameter slightly less than the test cell for snug fit. A water bath is used to control the temperature ± 0.5°C. in the test cell.

4.5.9.2 Procedure. Two hundred ml. of 0.01 percent test solution are placed in the test cell and allowed to equilibrate in a 70° C. bath. The stirrer is centered between the four baffles and clamped in position so that the liquid level is mid-way between the two blades on the stirrer. The stirring motor is run for 3 minutes at high speed (19,000 r.p.m) to allow the foam to reach maximum height. The stirrer is stopped at 3 minutes and foam height measured in centimeters after 5 seconds. The initial measurement is made by measuring the combined liquid-foam level from the top to the bottom baffle ring. All other measurements are foam height only.

4.5.9.3 Reporting results. Two tests are run with each solution and the average of both five second results is the reported value.

4.5.10 Moisture content. Determine the moisture content in accordance with sections 50 to 53 of ASTM D 2022-64.

4.5.11 Fineness.

4.5.11.1 Apparatus. Test shall be made using a single-eccentric type mechanical shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of one strike per minute. Sieves shall conform to RR-S-366.

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4.5.11.2 Procedure. Place a 100 g. sample on coarser screen of the nest of appropriate sieves and screen in shaking apparatus. Continue screening for 5 minute intervals until the weight retained on finer screen is reduced by not more than 0.1 g. during successive intervals. Weigh material on finer screen and calculate percent of material falling within the appropriate range.

NOTE. In order to evaluate results properly, excessive humidities should be avoided. In the event of failure, retests shall be conducted at a relative humidity not exceeding 50 percent.

4.5.12 Corrosion. The corrosion test shall be performed as follows: Polish by buffing two strips 1 cm. wide by 8 cm. long, of each of the following corrosion-resisting steels; classes 302, 304, and 430, condition A, finished No. 4, conforming to QQ-S-766. Degrease the strips by swabbing in a mixture of equal parts of alcohol and acetone. Allow the solvent to evaporate, heat in an oven at 105° C. for 30 minutes, cool in a desiccator. Remove the strips from the desiccator and quickly weigh them to the nearest 0.1 mg. Into test tubes measuring 32 mm. by 300 mm., pour 160 ml. of a mixture of 6.5 g. of the compound and 350 ml. of distilled water. Completely immerse the sample strips in the test solutions and maintain them at 175° F. for a period of 2 hours. At the end of 2 hours, remove the strips, rinse thoroughly with tap water, tho with acetone, and dry for 20 minutes in an oven at 100° to 105° C. Cool in a desiccator to room temperature. Remove the strips from the desiccator and quickly reweigh them to the nearest 0.1 mg., and examine for evidence of corrosion such as pitting, etching, or discoloration. Calculate the change of weight of strips as milligrams per square centimeter per hour of exposed surface.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Two pounds of dishwashing compound shall be packaged in a moisture-vaporproof package which can meet the water vapor permeability requirements of L-P-378 for Type I or Type II film.

5.1.2 Level B. The dishwashing compound shall be packaged as specified in 5.1.1.

5.1.3 Level C. Dishwashing compound shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Twenty-four two-pound packages shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed in accordance with the box specification.

5.2.1.2 Twenty-five pounds of dishwashing compound shall be packed in a fiber drum in accordance with PPP-P-45.

5.2.2 Level B

5.2.2.1 Twenty-four two-pound packages shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the box specification.

5.2.2.2 Twenty-five pounds of dishwashing compound shall be packed in a fiber drum in accordance with PPP-P-45 except that an alternative liner bag closure shall be by tightly twisting the mouth opening, folding over the twisted portion and applying a plastic coated wire tie around the center of the fold.

5.2.3 Level C. Dishwashing compound in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival destination in compliance with Uniform Freight Classification Committee or National Motor Freight Board, as applicable.

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5.3 Labeling and marking. In addition to the labeling requirements of 3.15 and any special markings required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking of shipping containers shall be in accordance with Fed. Std. No. 123 and shall contain the labeling data specified in 3.16.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, marking of shipping containers shall be in accordance with MIL-STD-129 and shall contain the labeling data specified in 3.16.

6. NOTES

6.1 Intended use. Chlorinated machine dishwashing compound described in this specification is intended to alleviate the problem of staining of chinaware. It is not intended for use as a destainer dip.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and level of packaging and packing required (see 5.1 and 5.2).
(c) Marking document required (see 5.3).

MILITARY INTEREST:

CUSTODIANS

Army - GL
Navy - SH
Air Force - 84

Review activities:

GL, MD
SH
Air Force - 84, 85

User activities:

MC, YD
26

Preparing activity:

GSA

CIVIL AGENCIES INTEREST:

AGR
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